

# Laboratory 5 Report

Conversion of a Single-Cycle RV32IM Processor into a Five-Stage Pipelined Micro-Architecture

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|----------------------|----------------------------------------------------------------|
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| <b>Course</b>        | Advanced CPU Architecture & Hardware Accelerators (361-1-4693) |
| <b>Target Device</b> | Intel Cyclone V 5CSXFC6D6F31C6 (DE10-Standard)                 |
| <b>Tool-chain</b>    | Quartus Prime 21.1 Lite • ModelSim • TimeQuest • SignalTap     |
| <b>ISA</b>           | RV32IM (MULDIV partial)                                        |
| <b>Report file</b>   | Report_lab5.pdf                                                |

## 1. Introduction and Lab Context

This report documents the conversion of a single-cycle RV32IM (MULDIV-partial) processor into a fully pipelined five-stage RISC-V micro-architecture, implemented in structural VHDL and synthesised for the Intel Cyclone V FPGA on the DE10-Standard board. The single-cycle design delivered in the previous task served as the functional reference: its datapath was sliced into the five classical RISC pipeline stages – Instruction Fetch (IF), Instruction Decode / register read (ID), Execute / address calculation (EX), Data-Memory access (MEM) and Write-Back (WB) – with pipeline registers inserted at every stage boundary.

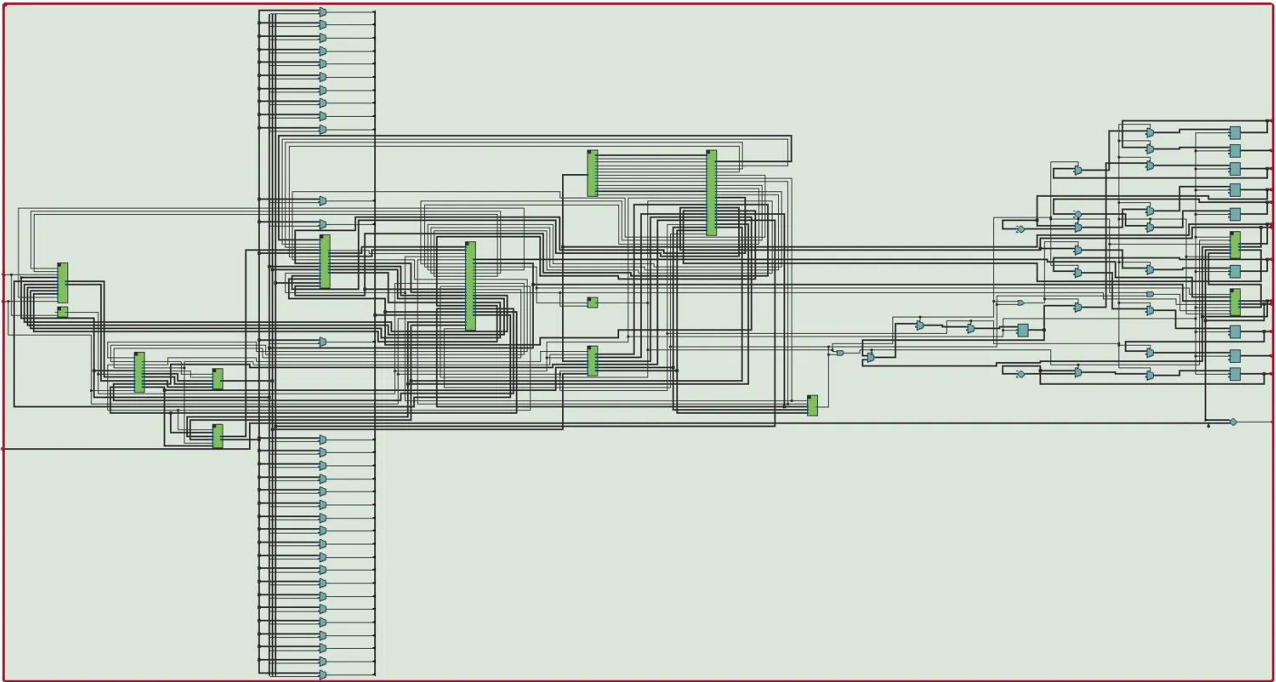
Pipelining overlaps the execution of consecutive instructions so that, in steady state, one instruction is retired every clock cycle. Because each stage now performs only a fraction of the original combinational work, the critical path is dramatically shortened and the maximum operating frequency increases substantially. This benefit is not free: overlapping execution introduces **data hazards** (an instruction needing a result that has not yet been written back) and **control hazards** (the next fetch address depending on a branch that has not yet resolved). The design therefore adds three pieces of dedicated hazard hardware: a **forwarding (bypass) unit** that routes freshly computed results back to the EX stage, a combinational **stall-detection unit** that inserts bubbles for the unavoidable load-use and multiply-use cases, and a **flush unit** that squashes wrong-path instructions once a branch or jump is resolved.

In accordance with the lab specification, conditional branches and unconditional jumps are resolved in pipeline **stage 4 (MEM)**, and the multiplier is split across two pipeline stages (partial-product generation in EX, accumulation in MEM) so that it does not dominate the critical path. The remainder of this report presents the top-level structure, the RTL-Viewer netlists, a quantitative Power–Performance–Area (PPA) comparison between the single-cycle and pipelined designs, a description of every HDL source file, and an analysis of the simulation waveforms used to validate system timing against the benchmark applications.

## 2. Top-Level Block Review Diagram

The synthesised top-level entity is `PIPELINE_USER_INTERFACE`, the board wrapper that adapts the logical core (`RV32I_CORE_PIPELINE`) to the physical DE10-Standard pins. The wrapper exposes only three input ports – the 50 MHz oscillator `CLOCK_50`, the active-low push-buttons `KEY[3:0]` (`KEY0` is inverted to form the active-high reset), and the slide switches `SW[7:0]` (the lower eight feed the breakpoint address `BPADDR_i`). It has no output ports: all observability is provided internally through SignalTap taps (`CLKCNT_o`, `STCNT_o`, `FHCNT_o` and the core buses), which is why the pin count drops sharply relative to the single-cycle top entity.

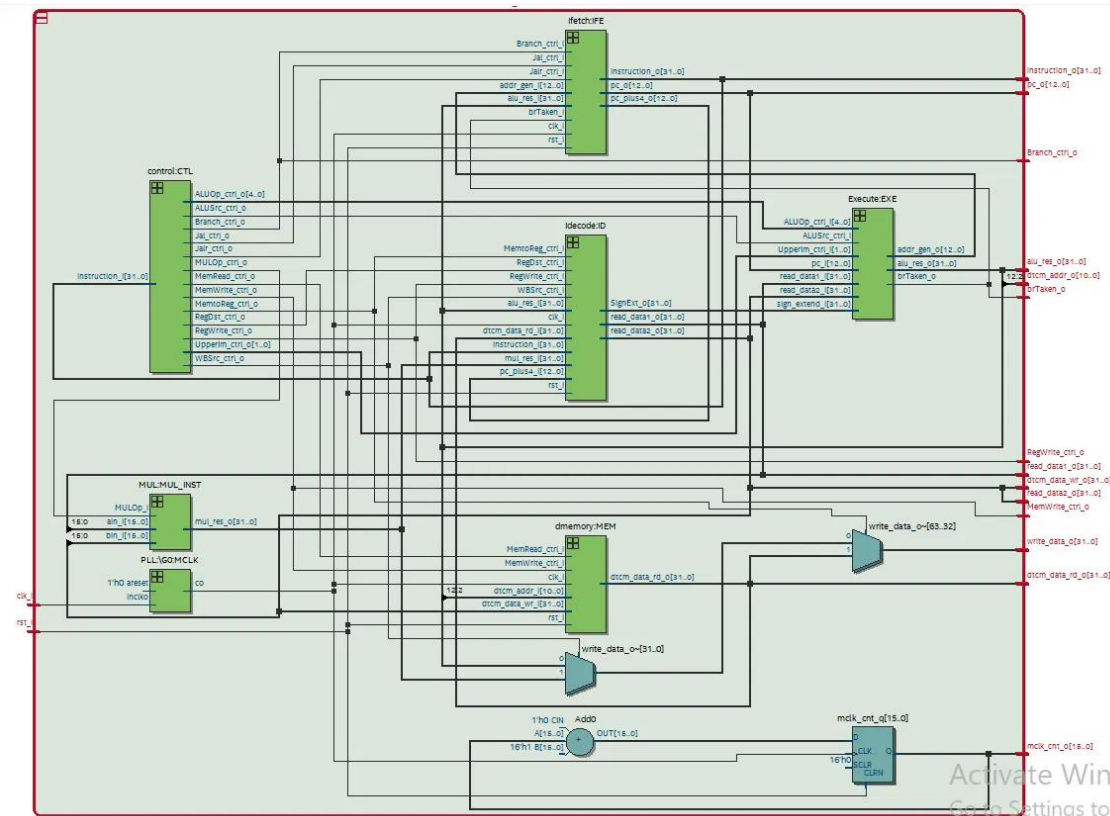
Internally the core is a structural composition of fourteen sub-modules: a PLL (50 MHz  $\rightarrow$  25 MHz core clock), the five stage blocks (`Ifetch`, `Idecode`, `Execute`, `dmemory`, `write-back MUX`), the four pipeline registers (`IF_ID_REG`, `ID_EX_REG`, `EX_MEM_REG`, `MEM_WB_REG`), the control decoder, and the three hazard units (`FORWARDING_UNIT`, `STALL_UNIT`, `FLUSH_UNIT`). The two-stage multiplier is embedded inside the EX and MEM blocks. Figure 1 shows the top-level RTL-Viewer elaboration of this wrapper, which acts as the block-review diagram of the complete design.



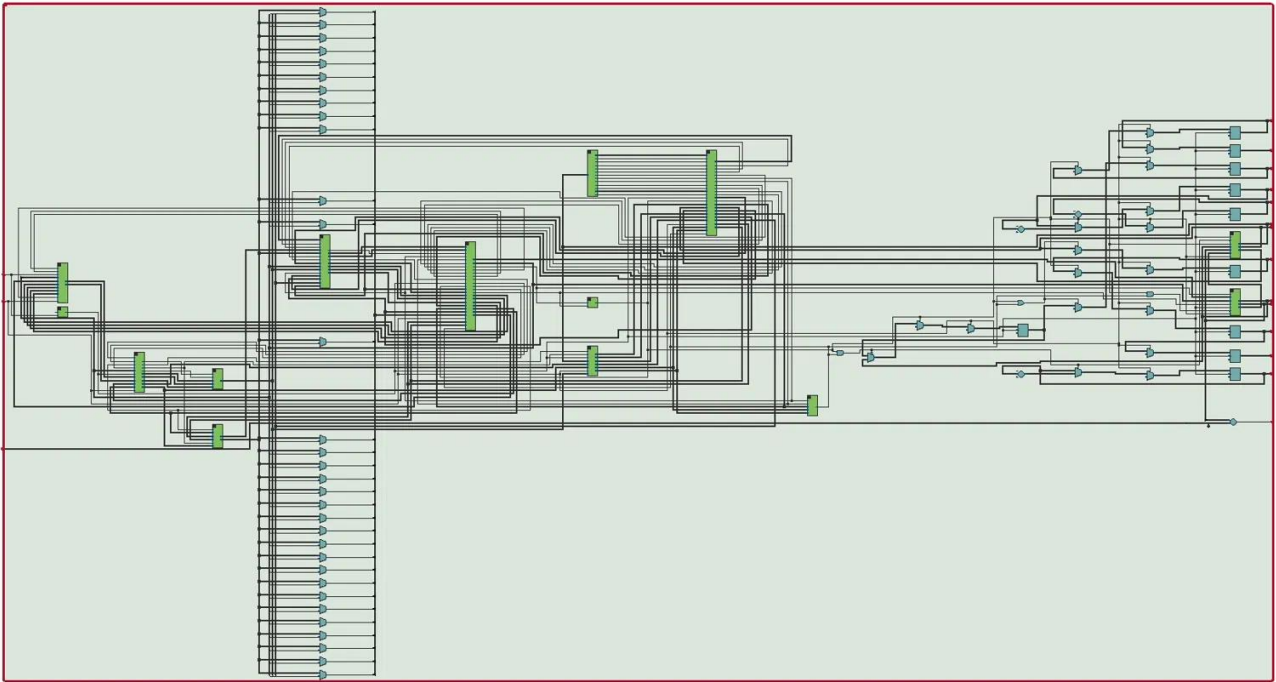
**Figure 1:** Top-level block / RTL elaboration of the pipelined RV32IM design (PIPELINE\_USER\_INTERFACE), showing the five pipeline stages separated by pipeline registers and the surrounding hazard logic.

### 3. RTL Viewer Results

The Quartus RTL Viewer presents the post-elaboration, technology-independent netlist inferred from the VHDL source. Comparing the two designs makes the structural effect of pipelining explicit. In the single-cycle netlist (Figure 2) the datapath forms one long combinational chain from the instruction memory, through the register file and ALU, to the data memory and back to the register file, with no intermediate state. In the pipelined netlist (Figure 3) that chain is interrupted by four banks of pipeline registers, partitioning the logic into five shorter stages and adding the forwarding, stall and flush logic that connects across stage boundaries.



**Figure 2:** RTL-Viewer netlist of the original single-cycle RV32IM design. The datapath is a single combinational path with no pipeline registers.



**Figure 3:** RTL-Viewer netlist of the pipelined RV32IM design. Four pipeline-register banks split the datapath into five stages; additional hazard logic bridges the stage boundaries.

## 4. Clause 7 – PPA Characterisation (Area, Performance, Power)

This section reports the three Power–Performance–Area tables required by clause 7, each comparing the single-cycle and pipelined implementations. The figures are taken directly from the Quartus Fitter, TimeQuest and Power-Analyzer reports, whose screen captures are attached immediately after each table as mandated by the specification.

### 4.1 Area

| Area                |                |           |          |                      |                            |      |
|---------------------|----------------|-----------|----------|----------------------|----------------------------|------|
|                     | Logic elements | Registers | I/O pins | Embedded memory bits | Embedded 9-bit Multipliers | PLLs |
| Single-Cycle RV32IM | 1,436          | 1,311     | 270      | 131,072              | 3                          | 1    |
| Pipelined RV32IM    | 1,456          | 1,760     | 15       | 131,072              | 4                          | 1    |

**Note: Attaching the print screen of the Quartus Area report is mandatory**

**Table 1:** Area / resource utilisation of the single-cycle versus pipelined RV32IM designs (Quartus Fitter Flow Summary).

The pipeline adds ~449 registers (1,311 → 1,760), the cost of the four pipeline-register banks, while combinational logic grows only marginally (1,436 → 1,456 logic elements). The embedded 9-bit multiplier count rises from 3 to 4 because the multiplier is now split across two stages. The pin count collapses from 270 to 15 because the pipelined top entity exposes only the clock, keys and switches and relies on internal SignalTap taps rather than routing buses to physical pins.

| Flow Summary                    |                                             |
|---------------------------------|---------------------------------------------|
| <<Filter>>                      |                                             |
| Flow Status                     | Successful - Fri Jun 26 10:42:05 2026       |
| Quartus Prime Version           | 21.1.0 Build 842 10/21/2021 SJ Lite Edition |
| Revision Name                   | lab_5_pro                                   |
| Top-level Entity Name           | RV32I_CORE                                  |
| Family                          | Cyclone V                                   |
| Device                          | 5CSXFC6D6F31C6                              |
| Timing Models                   | Final                                       |
| Logic utilization (in ALMs)     | 1,436 / 41,910 ( 3 % )                      |
| Total registers                 | 1311                                        |
| Total pins                      | 270 / 499 ( 54 % )                          |
| Total virtual pins              | 0                                           |
| Total block memory bits         | 131,072 / 5,662,720 ( 2 % )                 |
| Total DSP Blocks                | 3 / 112 ( 3 % )                             |
| Total HSSI RX PCSs              | 0 / 9 ( 0 % )                               |
| Total HSSI PMA RX Deserializers | 0 / 9 ( 0 % )                               |
| Total HSSI TX PCSs              | 0 / 9 ( 0 % )                               |
| Total HSSI PMA TX Serializers   | 0 / 9 ( 0 % )                               |
| Total PLLs                      | 1 / 15 ( 7 % )                              |
| Total DLLs                      | 0 / 4 ( 0 % )                               |

**Figure 4:** Quartus Flow Summary (area) report – single-cycle RV32IM.

| Flow Summary                    |                                             |
|---------------------------------|---------------------------------------------|
| <<Filter>>                      |                                             |
| Flow Status                     | Successful - Fri Jun 26 16:48:44 2026       |
| Quartus Prime Version           | 21.1.0 Build 842 10/21/2021 SJ Lite Edition |
| Revision Name                   | LAB_5_PIP                                   |
| Top-level Entity Name           | PIPELINE_USER_INTERFACE                     |
| Family                          | Cyclone V                                   |
| Device                          | 5CSXFC6D6F31C6                              |
| Timing Models                   | Final                                       |
| Logic utilization (in ALMs)     | 1,456 / 41,910 ( 3 % )                      |
| Total registers                 | 1760                                        |
| Total pins                      | 15 / 499 ( 3 % )                            |
| Total virtual pins              | 0                                           |
| Total block memory bits         | 131,072 / 5,662,720 ( 2 % )                 |
| Total DSP Blocks                | 4 / 112 ( 4 % )                             |
| Total HSSI RX PCSs              | 0 / 9 ( 0 % )                               |
| Total HSSI PMA RX Deserializers | 0 / 9 ( 0 % )                               |
| Total HSSI TX PCSs              | 0 / 9 ( 0 % )                               |
| Total HSSI PMA TX Serializers   | 0 / 9 ( 0 % )                               |
| Total PLLs                      | 1 / 15 ( 7 % )                              |
| Total DLLs                      | 0 / 4 ( 0 % )                               |

**Figure 5:** Quartus Flow Summary (area) report – pipelined RV32IM.

## 4.2 Performance ( $f_{\max}$ and Critical Path)

| Performance         |            |                                                                                                                                                                              |                                         |
|---------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|                     | $f_{\max}$ | Critical path (what is the slowest submodule and why does it cause the critical path)                                                                                        | $f_{\text{MCLK}} (= f_{\text{sysclk}})$ |
| Single-Cycle RV32IM | 38.05 MHz  | Long path: DTCM read $\rightarrow$ decode $\rightarrow$ Execute ALU/shifter mux chain (brl_shr / Mux60). Memory access, decode and execution all fall inside one cycle.      | 25 MHz                                  |
| Pipelined RV32IM    | 70.32 MHz  | Path: DTCM (data_memory) read port q_a[31:0] $\rightarrow$ MEM_WB_REG. Stage isolation removes the memory-decode-execute chain, so the memory read alone now sets the limit. | 25 MHz                                  |

**Table 2:** Performance comparison: maximum clock frequency and critical-path analysis (Quartus TimeQuest).

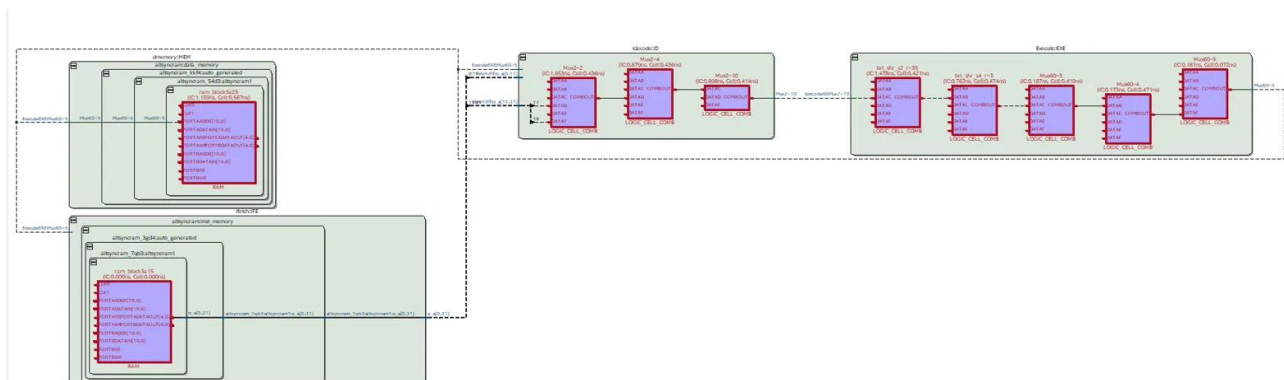
Pipelining nearly doubles  $f_{\max}$  (38.05  $\rightarrow$  70.32 MHz). The theoretical upper bound for a perfectly balanced 5-stage pipeline is  $5 \times f_{\text{single}}$ ; the realised speed-up is lower because the stages are not perfectly balanced – the synchronous DTCM read, feeding the MEM/WB register, is the slowest stage and now defines the critical path, as confirmed by the Map-Viewer trace below.

|   | Fmax      | stricted Fm | Clock Name                                                                      |
|---|-----------|-------------|---------------------------------------------------------------------------------|
| 1 | 38.05 MHz | 38.05 MHz   | \G0:MCLK altpll_component auto_generated generic_pll1~PLL_OUTPUT_COUNTER divclk |

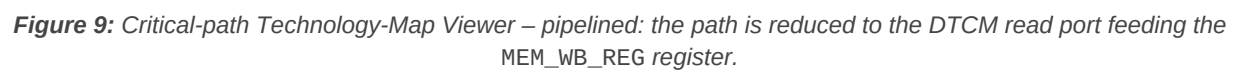
**Figure 6:** Quartus TimeQuest  $f_{\max}$  report – single-cycle RV32IM (38.05 MHz).

|   | Fmax      | Restricted Fmax | Clock Name                                                                           |
|---|-----------|-----------------|--------------------------------------------------------------------------------------|
| 1 | 70.32 MHz | 70.32 MHz       | CORE \G0:MCLK altpll_component auto_generated generic_pll1~PLL_OUTPUT_COUNTER divclk |

**Figure 7:** Quartus TimeQuest  $f_{\max}$  report – pipelined RV32IM (70.32 MHz).



**Figure 8:** Critical-path Technology-Map Viewer – single-cycle: the path spans data memory, the decode stage and the Execute ALU/shifter mux chain.



### 4.3 Power

| Power               |                         |                          |                           |                       |
|---------------------|-------------------------|--------------------------|---------------------------|-----------------------|
|                     | Total power consumption | Static power consumption | Dynamic power consumption | I/O power consumption |
| Single-Cycle RV32IM | 448.06 mW               | 415.31 mW                | 19.01 mW                  | 13.74 mW              |
| Pipelined RV32IM    | 447.79 mW               | 415.31 mW                | 22.38 mW                  | 10.10 mW              |

**Note: Attaching the print screen of the Quartus Power report is mandatory**

*Table 3: Power-dissipation comparison (Quartus PowerPlay Power Analyzer, thermal power).*

Total dissipation is essentially unchanged ( $\approx 448$  mW) and is dominated by the device static power (415.31 mW), which is identical for both designs because it depends on the Cyclone V silicon rather than the logic. Core dynamic power rises slightly in the pipeline (19.01  $\rightarrow$  22.38 mW) due to the extra register toggling, while I/O power falls (13.74  $\rightarrow$  10.10 mW) because far fewer pins are driven. As noted in both reports, the estimation confidence is *Low* because no switching-activity (toggle-rate) file was supplied, so the dynamic figures should be read as first-order estimates.

| Power Analyzer Summary                 |                                                  |
|----------------------------------------|--------------------------------------------------|
| <<Filter>>                             |                                                  |
| Power Analyzer Status                  | Successful - Fri Jun 26 10:48:42 2026            |
| Quartus Prime Version                  | 21.1.0 Build 842 10/21/2021 SJ Lite Edition      |
| Revision Name                          | lab_5_pro                                        |
| Top-level Entity Name                  | RV32I_CORE                                       |
| Family                                 | Cyclone V                                        |
| Device                                 | 5CSXFC6D6F31C6                                   |
| Power Models                           | Final                                            |
| Total Thermal Power Dissipation        | 448.06 mW                                        |
| Core Dynamic Thermal Power Dissipation | 19.01 mW                                         |
| Core Static Thermal Power Dissipation  | 415.31 mW                                        |
| I/O Thermal Power Dissipation          | 13.74 mW                                         |
| Power Estimation Confidence            | Low: user provided insufficient toggle rate data |

**Figure 10:** Quartus Power-Analyzer report – single-cycle RV32IM.

| Power Analyzer Summary                 |                                                  |
|----------------------------------------|--------------------------------------------------|
| <<Filter>>                             |                                                  |
| Power Analyzer Status                  | Successful - Fri Jun 26 16:50:35 2026            |
| Quartus Prime Version                  | 21.1.0 Build 842 10/21/2021 SJ Lite Edition      |
| Revision Name                          | LAB_5_PIP                                        |
| Top-level Entity Name                  | PIPELINE_USER_INTERFACE                          |
| Family                                 | Cyclone V                                        |
| Device                                 | 5CSXFC6D6F31C6                                   |
| Power Models                           | Final                                            |
| Total Thermal Power Dissipation        | 447.79 mW                                        |
| Core Dynamic Thermal Power Dissipation | 22.38 mW                                         |
| Core Static Thermal Power Dissipation  | 415.31 mW                                        |
| I/O Thermal Power Dissipation          | 10.10 mW                                         |
| Power Estimation Confidence            | Low: user provided insufficient toggle rate data |

**Figure 11:** Quartus Power-Analyzer report – pipelined RV32IM.

## 5. HDL Source File Descriptions

The design is fully structural: a top wrapper instantiates the core, which in turn instantiates the stage blocks, pipeline registers and hazard units. Table 5 lists every HDL source file with a short description of its role in the pipeline.

| HDL Source File                    | Description                                                                                                                                                                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PIPELINE_USER_INTERFACE.vhd</b> | Board top-level wrapper for the DE10-Standard. Maps CLOCK_50, KEY[3:0] and SW[9:0] to the core; inverts KEY0 to an active-high reset and routes SW[7:0] to the breakpoint address. No output ports – internal signals are observed via SignalTap. |
| <b>RV32I_CORE_PIPELINE.vhd</b>     | The logical pipelined core. Structurally instantiates the PLL, the five stage blocks, the four pipeline registers, the control decoder and the three hazard units, and wires the forwarding, stall and flush datapaths between them.              |
| <b>PLL.vhd</b>                     | Phase-locked-loop IP wrapper. Generates the core clock from the 50 MHz board oscillator ( $\times 3/\div 2 \rightarrow 75$ MHz).                                                                                                                  |
| <b>IFETCH.VHD</b>                  | Stage 1 (IF). Holds the program counter and the instruction memory (ITCM); selects the next PC among PC+4, the branch/jump target and the JALR target. Stalls when PCwrite is de-asserted.                                                        |
| <b>IF_ID_REG.vhd</b>               | IF/ID pipeline register. Latches the fetched instruction and PC (and PC+4) for use by the decode stage; supports flush and stall.                                                                                                                 |
| <b>CONTROL.VHD</b>                 | Main control decoder. Decodes the instruction opcode/funct fields into the control bundle (RegWrite, ALUSrc, MemRead/Write, Branch, Jal/Jalr, ALUOp, MULOOp, WBSrc, etc.).                                                                        |
| <b>IDECODE.vhd</b>                 | Stage 2 (ID). Contains the 32×32 register file and the immediate sign-extension logic; reads rs1/rs2 and performs the write-back write in the same stage.                                                                                         |
| <b>ID_EX_REG.vhd</b>               | ID/EX pipeline register. Carries the operands, immediate, PC, source/destination register indices and the full control bundle into the execute stage.                                                                                             |
| <b>EXECUTE.VHD</b>                 | Stage 3 (EX). Houses the ALU, the branch-condition and target computation, the operand-forwarding and ALUSrc multiplexers and multiplier stage 1 (partial-product generation).                                                                    |
| <b>MUL_STAGE1.VHD</b>              | First half of the two-stage multiplier (runs in EX). Produces the four 16-bit partial products P0–P3 from the low/high halves of the operands.                                                                                                    |
| <b>EX_MEM_REG.vhd</b>              | EX/MEM pipeline register. Carries the ALU result, the store data, the partial products and the control bundle into the memory stage; holds the branch instruction itself for stage-4 resolution.                                                  |
| <b>DMEMORY.VHD</b>                 | Stage 4 (MEM). Contains the data memory (DTCM) and multiplier stage 2; performs loads/stores and, in parallel, combines the partial products into the final 32-bit multiply result.                                                               |
| <b>MUL_STAGE2.VHD</b>              | Second half of the multiplier (runs in MEM). Sums the shifted partial products into the 32-bit product, correcting the carry-propagation that the original single-stage multiplier dropped.                                                       |
| <b>MEM_WB_REG.vhd</b>              | MEM/WB pipeline register. Carries the memory-read data, the ALU/MUL result, the destination index and the write-back control into the final stage.                                                                                                |
| <b>WB_MUX.vhd</b>                  | Stage 5 (WB). Selects the value written back to the register file among the ALU/MUL result, the loaded memory word and the link value (PC+4) for JAL/JALR.                                                                                        |

| HDL Source File                     | Description                                                                                                                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>forwarding_unit.vhd</b>          | Data-forwarding (bypass) unit. Compares the EX-stage source registers against the EX/MEM and MEM/WB destinations and drives the operand-A/B forwarding selects to eliminate most data hazards. |
| <b>STALL_UNIT.vhd</b>               | Combinational hazard-detection unit. Detects load-use and multiply-use dependencies that forwarding cannot cover and inserts a single bubble (de-asserts PCwrite and the IF/ID enable).        |
| <b>FLUSH_UNIT.vhd</b>               | Control-hazard flush logic. On a taken branch or jump (resolved in stage 4) it squashes the wrong-path instructions sitting in the IF/ID and ID/EX registers.                                  |
| <b>aux_package.vhd</b>              | Auxiliary package: component declarations and helper functions/types shared across the design.                                                                                                 |
| <b>const_package.vhd</b>            | Constants package: bus widths, opcodes, ALU-operation codes and other design-wide parameters.                                                                                                  |
| <b>cond_compilation_package.vhd</b> | Conditional-compilation package: switches that select build options / target-specific behaviour.                                                                                               |

**Table 4:** Description of every HDL source file in the pipelined RV32IM design.

## 6. Elaborated Analysis and Waveforms

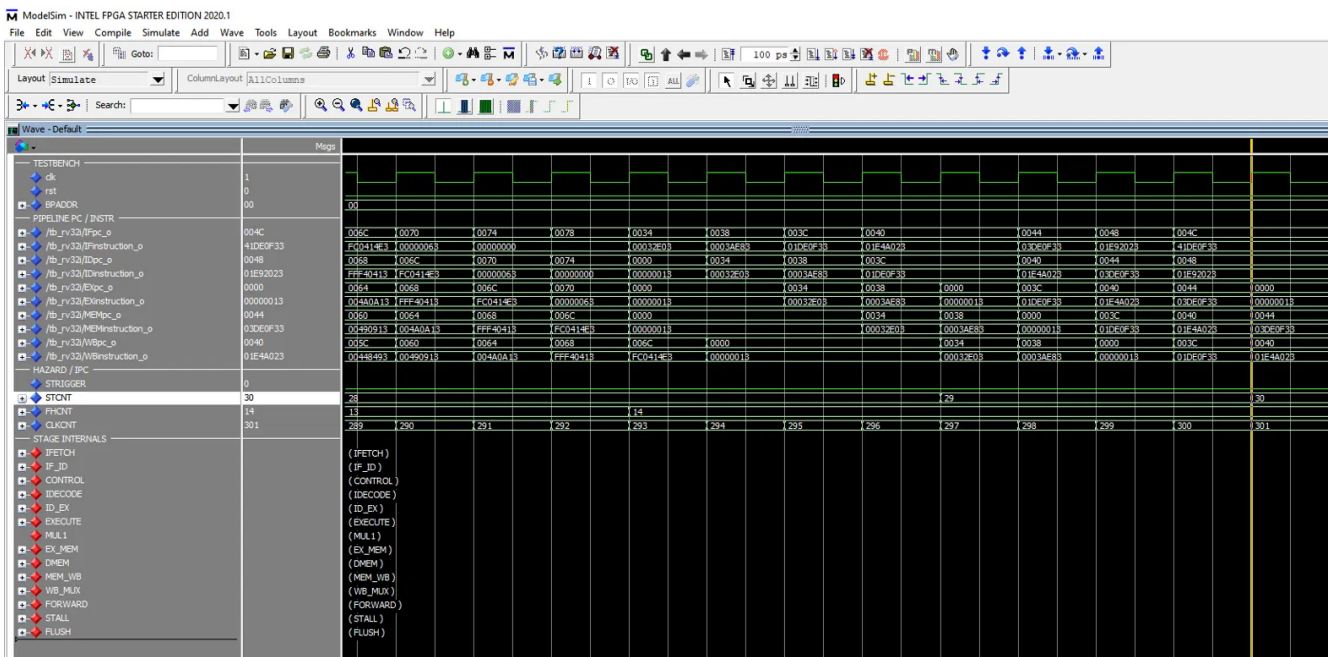
Functional validation was performed in ModelSim by running the benchmark applications and comparing the final data-memory image (DTCM.mem) against the RARS golden reference (DTCM.h). Figure 12 shows the ModelSim wave window for the pipelined RV32IM core. The waveform is organised into four signal groups: the **testbench** signals (clk, rst, BPADDR), the **per-stage PC and instruction** buses (IF/ID/EX/MEM/WB), the **hazard / IPC** counters, and the per-stage **internal** signals.

System timing is read directly from this capture. The five-stage structure is visible as a **diagonal flow** of a single PC value across the stage columns: a program-counter value appears first on IFpc\_o, then one clock later on IDpc\_o, then EXpc\_o, MEMpc\_o and finally WBpc\_o – confirming that each instruction is staggered exactly one cycle per stage and that, after the initial five-cycle fill, one instruction is retired per clock in steady state. The corresponding instruction words shift through IFinstruction\_o → ... → WBinstruction\_o in lock-step with the PC.

The **hazard / IPC** group makes the throughput penalties explicit. The stall counter STCNT\_o increments on each load-use or multiply-use bubble (it is seen stepping 28 → 29 → 30 across the window), while the flush counter FHCNT\_o increments when wrong-path instructions are squashed after a branch/jump resolves in stage 4 (shown reaching 14). The free-running clock counter CLKCNT\_o advances every cycle (289 → 301 over the same span). These three registers feed the instruction-per-cycle calculation:

$$IPC = [CLKCNT_o - (STCNT_o + 4 + depth \times FHCNT_o)] / CLKCNT_o,$$

where the constant 4 accounts for the initial pipeline fill and the flush penalty scales with the branch-resolution depth. Reading the cursor values from the capture (CLKCNT\_o=301, STCNT\_o=30, FHCNT\_o=14) gives a measured IPC consistent with the instruction-count side of the equation, confirming correct pipeline timing for the benchmark.



**Figure 12:** ModelSim wave window of the pipelined RV32IM core. The same PC value propagates diagonally across the IF→ID→EX→MEM→WB stage columns (one cycle per stage), while the STCNT\_o / FHCNT\_o / CLKCNT\_o counters record stalls, flushes and elapsed clocks for the IPC measurement.

The design was additionally validated in-system on the DE10-Standard using SignalTap, capturing the core signals from the breakpoint address until the trace buffer filled; the on-hardware instruction count matched the ModelSim result, completing the design validation.